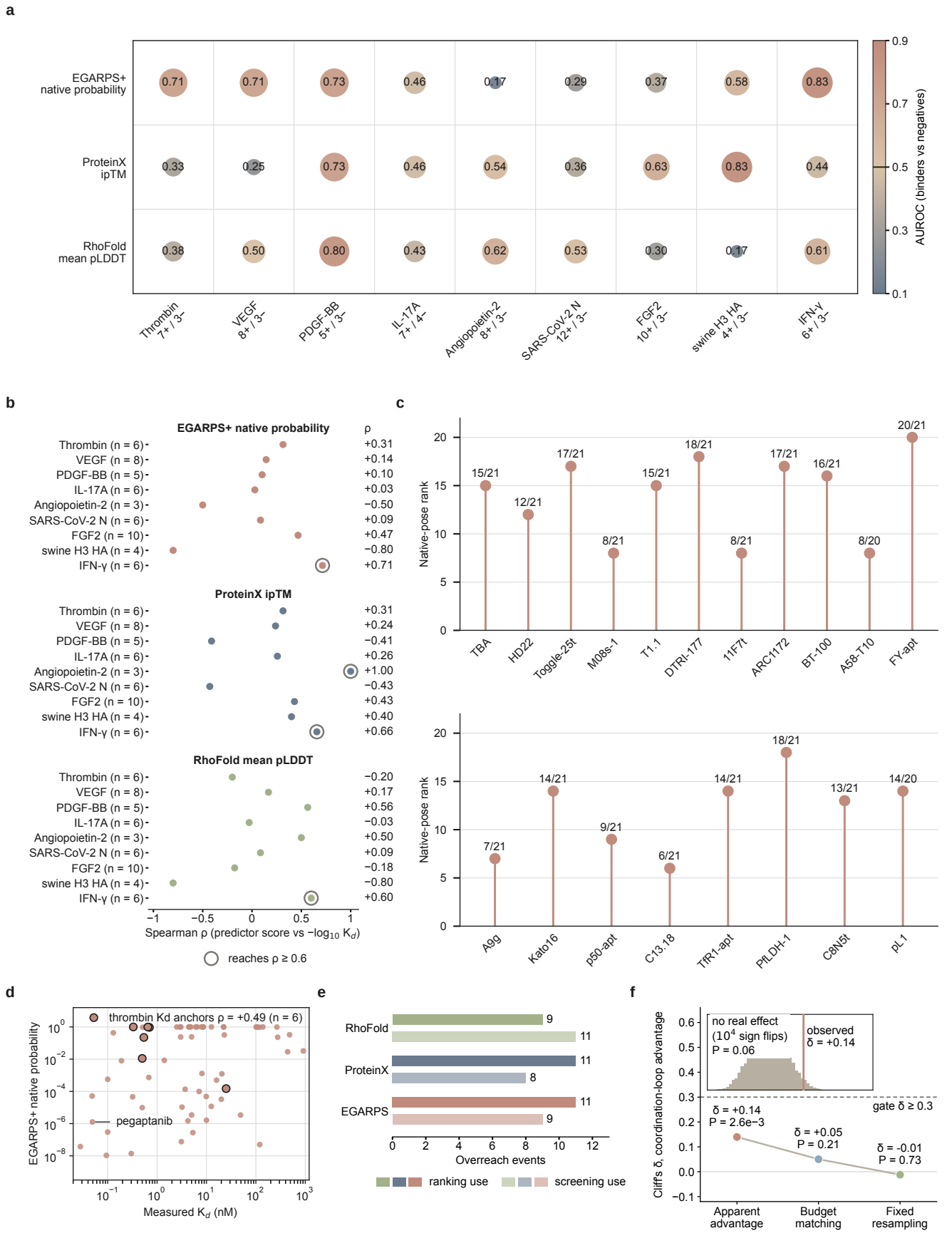

a
EGARPS+ native probability
ProteinX ipTM
RhoFold mean pLDDT
0.71 0.71 0.73 0.46 0.17 0.29 0.37 0.58 0.83
0.33 0.25 0.73 0.46 0.54 0.36 0.63 0.83 0.44
0.38 0.50 0.80 0.43 0.62 0.53 0.30 0.17 0.61
Thrombin 7+ / 3–
VEGF 8+ / 3–
PDGF-BB 5+ / 3–
IL-17A 7+ / 4–
Angiopoietin-2 8+ / 3–
SARS-CoV-2 N 12+ / 3–
FGF2 10+ / 3–
swine H3 HA 4+ / 3–
IFN-γ 6+ / 3–
AUROC (binders vs negatives)
0.9
0.7
0.5
0.3
0.1
b
EGARPS+ native probability
ρ
Thrombin (n = 6) +0.31
VEGF (n = 8) +0.14
PDGF-BB (n = 5) +0.10
IL-17A (n = 6) +0.03
Angiopoietin-2 (n = 3) −0.50
SARS-CoV-2 N (n = 6) +0.09
FGF2 (n = 10) +0.47
swine H3 HA (n = 4) −0.80
IFN-γ (n = 6) +0.71
ProteinX ipTM
Thrombin (n = 6) +0.31
VEGF (n = 8) +0.24
PDGF-BB (n = 5) −0.41
IL-17A (n = 6) +0.26
Angiopoietin-2 (n = 3) +1.00
SARS-CoV-2 N (n = 6) −0.43
FGF2 (n = 10) +0.43
swine H3 HA (n = 4) +0.40
IFN-γ (n = 6) +0.66
RhoFold mean pLDDT
Thrombin (n = 6) −0.20
VEGF (n = 8) +0.17
PDGF-BB (n = 5) +0.56
IL-17A (n = 6) −0.03
Angiopoietin-2 (n = 3) +0.50
SARS-CoV-2 N (n = 6) +0.09
FGF2 (n = 10) −0.18
swine H3 HA (n = 4) −0.80
IFN-γ (n = 6) +0.60
−1 −0.5 0 0.5 1
Spearman ρ (predictor score vs −log10 Kd)
reaches ρ ≥ 0.6
c
Native-pose rank
15/21 12/21 17/21 8/21 15/21 18/21 8/21 17/21 16/21 8/20 20/21
TBA HD22 Toggle-25t M08s-1 T1.1 DTRI-177 11F7t ARC1172 BT-100 A58-T10 FY-apt
7/21 14/21 9/21 6/21 14/21 18/21 13/21 14/20
A9g Kato16 p50-apt C13.18 TfR1-apt PfLDH-1 C8N5t pL1
d
EGARPS+ native probability
thrombin Kd anchors ρ = +0.49 (n = 6)
pegaptanib
Measured Kd (nM)
e
RhoFold 9 11
ProteinX 11 8
EGARPS 11 9
Overreach events
ranking use
screening use
f
Cliff's δ, coordination-loop advantage
no real effect
(10^4 sign flips)
P = 0.06
observed
δ = +0.14
gate δ ≥ 0.3
δ = +0.14
P = 2.6e−3
δ = +0.05
P = 0.21
δ = -0.01
P = 0.73
Apparent advantage
Budget matching
Fixed resampling